Dart Tutorial 2: Variables and Data Types

This chapter explains how to store and manage data in Dart using variables and data types. After completing this tutorial, students will understand how to declare variables, the different data types in Dart, and when to use each one.

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1. **Introduction to Variables**

A **variable** is a named container used to store data in memory so it can be used later in a program.

In Dart, every variable has a **type**, which defines what kind of data it can hold.

Variables make it possible to store, change, and reuse data throughout a program.

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2. **Why We Use Variables in Dart**

In real world apps, we constantly need to store information such as a user's name, age, or score.

Variables allow us to:

- Store data temporarily in memory.
- Reuse the same data multiple times.
- Update data dynamically as the program runs.
- Make programs flexible instead of hardcoding values.

Because of this, variables are the building blocks of every Dart program.

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3. **Declaring Variables in Dart**

Dart provides multiple ways to declare a variable.

```
var name = 'Ishfaq';
String city = 'Lahore';
int age = 22;
```

In this example:

- `var` lets Dart automatically detect the data type.
- `String` and `int` explicitly define the data type.

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4. var vs Explicit Data Types

Feature	var	Explicit Type
Type Detection	Automatic	Manual
Readability	Less clear	More clear
Flexibility	Type fixed after assignment	Type fixed at declaration
Recommended For	Quick variables	Production code

Although `var` is convenient, using explicit data types makes code easier to read and maintain.

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5. Common Data Types in Dart

Dart provides several built in data types to store different kinds of values.

Data Type	Description	Example
int	Whole numbers	10, 25, 100
double	Decimal numbers	3.14, 9.5
String	Text values	"Hello", 'Dart'
bool	True or False	true, false
dynamic	Can hold any type	"text" or 10

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6. Working with int and double

```
int marks = 90;
double percentage = 95.5;

print(marks);
print(percentage);
```

Output:

```
90
95.5
```

`int` is used for whole numbers, while `double` is used when decimal precision is required.

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7. **Working with String**

```
String course = 'Flutter Development';
print(course);
```

Strings can be written using single quotes `' '` or double quotes `" "`.

String Interpolation

```
String name = 'Ishfaq';
print('Welcome, $name');
```

Output:

```
Welcome, Ishfaq
```

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8. **Working with bool**

```
bool isLoggedIn = true;
print(isLoggedIn);
```

Output:

```
true
```

`bool` is mainly used for conditions and decision making in a program.

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9. **Understanding dynamic**

```
dynamic value = 'Hello';
value = 10;
print(value);
```

Output:

```
10
```

Unlike other types, a `dynamic` variable can change its data type during the program. It should be used carefully since it removes type safety.

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10. **Constants in Dart: final and const**

Sometimes we need values that should never change.

```
final String appName = 'Code With Ishfaq';
const double pi = 3.14;
```

Keyword	Value Set At	Can Be Changed
final	Run time	No
const	Compile time	No

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11. **Conclusion of Tutorial 2**

In this tutorial, students have learned:

1. What variables are and why they are used.
2. The difference between var and explicit data types.
3. The most common data types in Dart.
4. How to work with int, double, String, bool, and dynamic.
5. How to use final and const for fixed values.

In the next tutorial, we will learn about Operators in Dart.

This completes Dart Tutorial 2.